AER 1515 – Perception for Robotics

Lecture 3 – Optimization

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Optimization Improvements

- Jacobian and multi-dimensional problems
- Multi-objective optimization
- A couple of related examples for quadratic and NLP optimizations
- A conversion of a nonconvex problem to a convex problem for robotics
- Some example with Gaussians being optimized (MAP, ML)
- More detail on Lagrange multipliers and constrained solutions
- Connect to machine learning lecture

Outline

- Optimality and Unconstrained Optimization
- Conditions for optimality
- Optimization problem types
- Some solution methods

Unconstrained Optimization

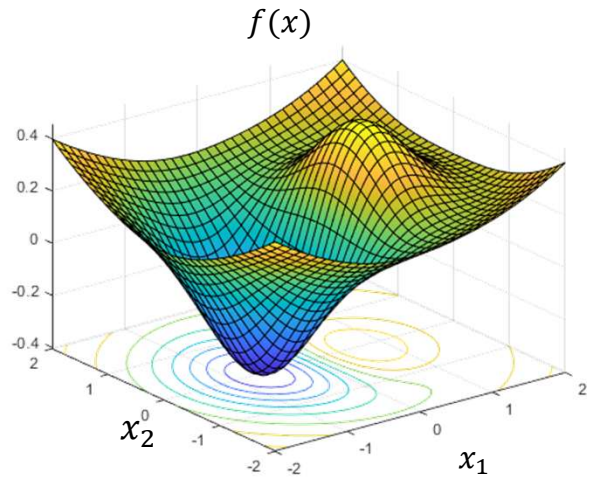
- Given a function that maps a vector of variables to the reals

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

- Find the minimum (or maximum) values of $f(x)$

$$\min_{x \in \mathbb{R}^n} \quad f(x)$$

  - Must map to a single value that ranks solutions and defines optimality

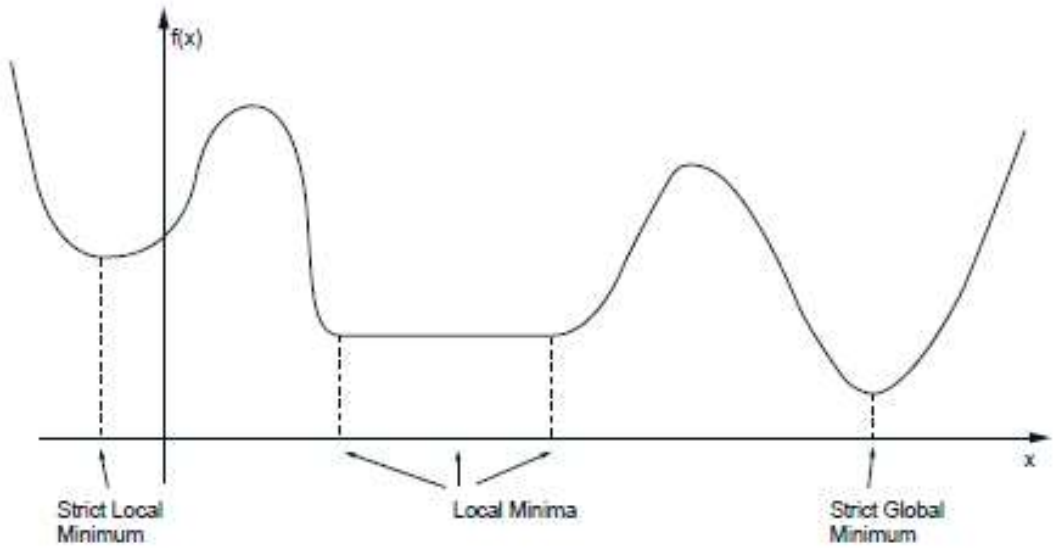


- The gradient and Hessian of the function $f(x)$ are usually assumed to exist, requires some smoothness
  - Gradient: $\nabla f(x) = \dfrac{\partial f}{\partial x}$
  - Hessian $\nabla^2 f(x) = \dfrac{\partial^2 f}{\partial x^2}$

Unconstrained Optimization

- Minima:
  - Local minimum, x^*:

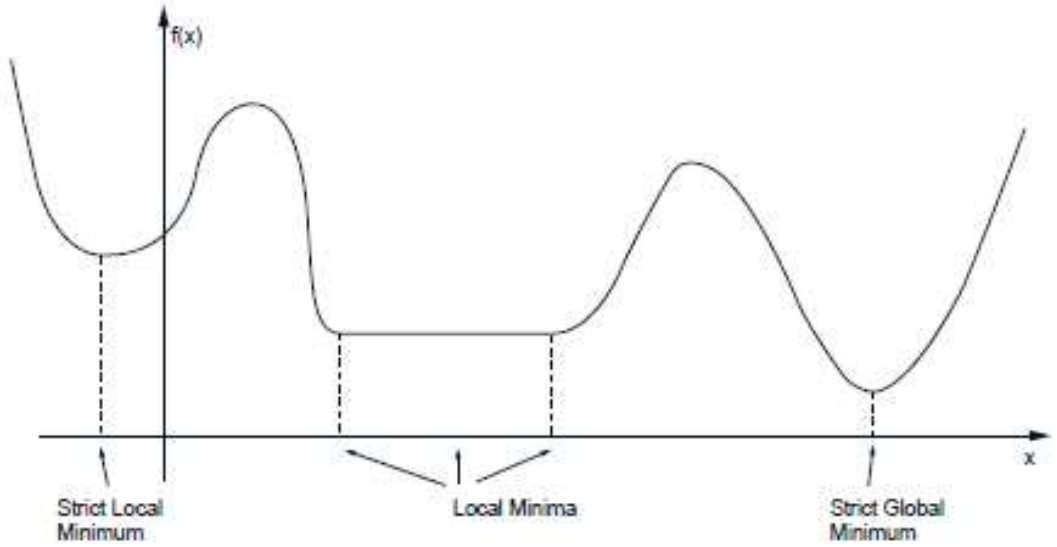
There exists an $\varepsilon > 0$ such that $f(x^*) \leq f(x)$, for all x with $||x - x^*|| < \varepsilon$



Unconstrained Optimization

- Minima:
  - Global minimum, x^*

$$f(x^*) \leq f(x), \text{ for all } x \in \mathbb{R}^n$$

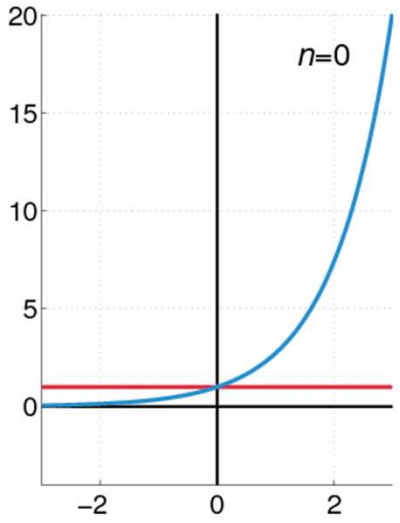


Conditions For Optimality

- For differentiable cost functions, can perform Taylor series expansion to find optimality conditions

- Taylor series of $f(x)$ about x

$$f(x+\Delta x) = f(x) + \nabla f(x)^T \Delta x + \frac{1}{2}\Delta x^T \nabla^2 f(x) \Delta x + H.O.T.$$



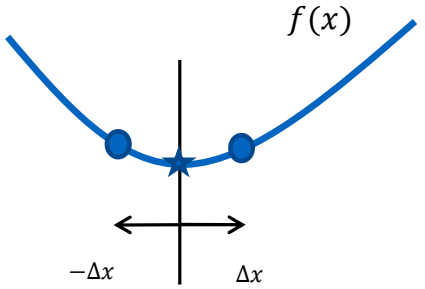
Conditions For Optimality

- Necessary conditions (NC)
  - If x^* is a local minimum, difference between minimum and nearby point should be non-negative by definition

$$f(x^* + \Delta x) - f(x^*) \geq 0$$

  - Similarly, for a negative step in x, the difference should be non-negative

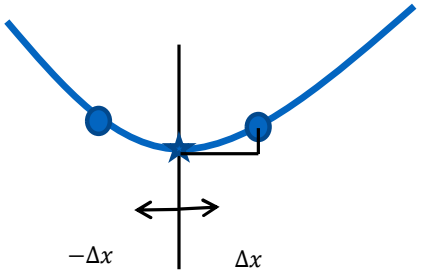
$$f(x^* - \Delta x) - f(x^*) \geq 0$$



Conditions For Optimality

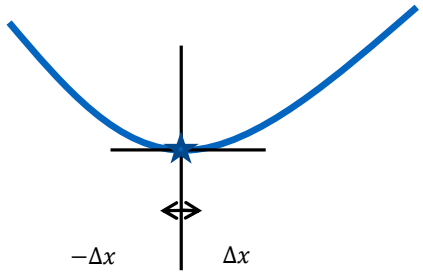
- Necessary conditions (NC)
  - As Δx->0, higher order terms in Taylor series disappear

$$f(x+\Delta x) - f(x) \approx \nabla f(x)^T \Delta x$$



  - First order term must satisfy above for Δx AND $-\Delta x$ in each element of x

$$\nabla f(x^*)^T \Delta x \geq 0 \text{ and } \nabla f(x^*)^T (-\Delta x) \geq 0$$



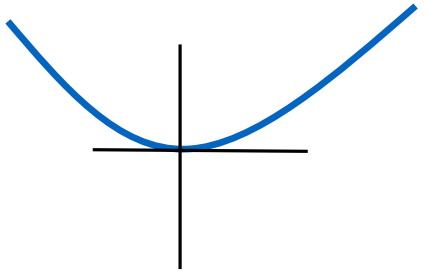
  - Necessary condition for optimality

$$\nabla f(x^*)^T = 0$$

Conditions for Optimality

- Sufficient conditions
  - Of all points that satisfy necessary conditions for optimality, which ones are truly local minima?
  - For all small excursions from optimal solution, cost increases
    - Since $\nabla f(x^*)^T = 0$
    - This means

$$f(x + \Delta x) - f(x) \approx \frac{1}{2}\Delta x^T \nabla^2 f(x) \Delta x > 0$$



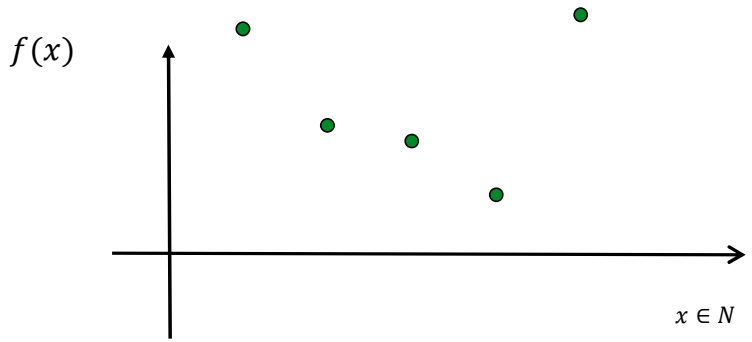
    - And so the sufficient condition for x^* to be a local minimum is

$$\nabla^2 f(x^*) \text{ is positive definite}$$

Optimality Conditions

- Note that these conditions are only useful if the gradient and Hessian exist

- Otherwise, resort to initial definition of optimality and demonstrate directly
  - Integer optimization



Constrained Optimization

- Standard form:

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{subject to} \quad & g(x) \leq 0 \\ & h(x) = 0 \end{aligned}$$

  - where

$$X \text{ can be any type of set, } f: X \to \mathbb{R}, g: X \to \mathbb{R}^p, h: X \to \mathbb{R}^q$$

- Specific classes of problems, depending on definitions of X, f, g, h.
- Very specific optimization engines, for every shade of problem

Lagrangian and Lagrange Multipliers

- By introducing Lagrange multipliers, can convert constrained problem to an unconstrained problem
  - Can directly apply unconstrained optimization technique to Lagrangian

$$L(x) = f(x) + \lambda^T g(x) + \nu^T h(x) \qquad \lambda \geq 0, \nu \neq 0$$

  - Results in expanded necessary and sufficient conditions for optimality

- In practice, best optimization algorithms treat constraints in a few different ways

Constrained Optimization

- Equality Constraints
  - Must be active, Lagrange multipliers must be non-zero

- Inequality Constraints (Karush, Kuhn, Tucker)
  - If active, Lagrange multipliers are non-negative
  - If inactive, Lagrange multipliers are zero

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0$$

$$\lambda_i^* \geq 0, i = 1, \ldots, m$$

$$\lambda_i^* = 0, i \notin A(x^*)$$

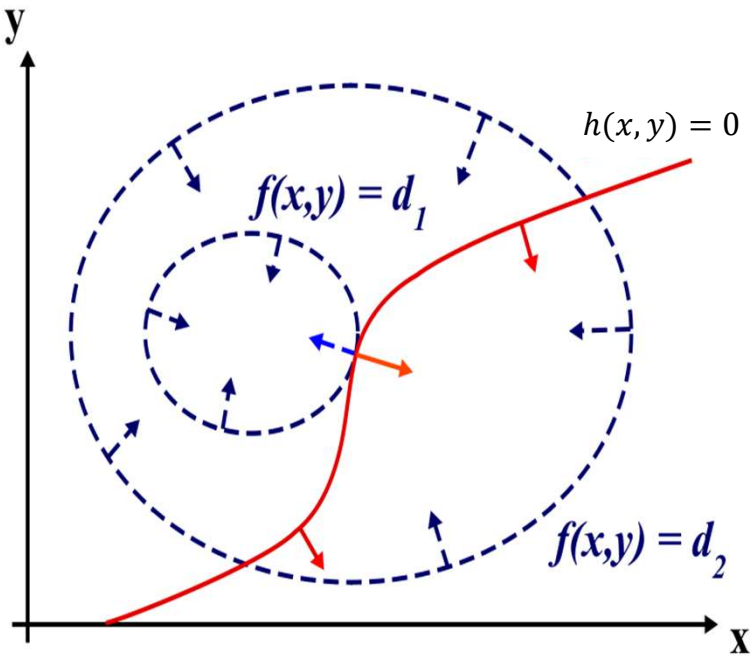
  - $A(x^*)$ is the set of active constraints

Constrained Minima

- If no active constraints, reverts to unconstrained problem
- For active constraints
  - Gradient of cost must be perpendicular to active constraint at minimum
    - Otherwise moving along constraint would reduce cost and remain feasible
  - Can be expressed as

$$\nabla f(x^*, y^*) + \nu^T \nabla h(x^*, y^*) = 0$$

    - with Lagrange multiplier $\nu \neq 0$



Lagrangian Duality

- We define the Lagrangian dual function as the infimum of the Lagrangian over x

$$d(\lambda, \nu) = \inf_{x \in X} L(x, \lambda, \nu) = \inf_{x \in X} (f(x) + \lambda^T g(x) + \nu^T h(x))$$

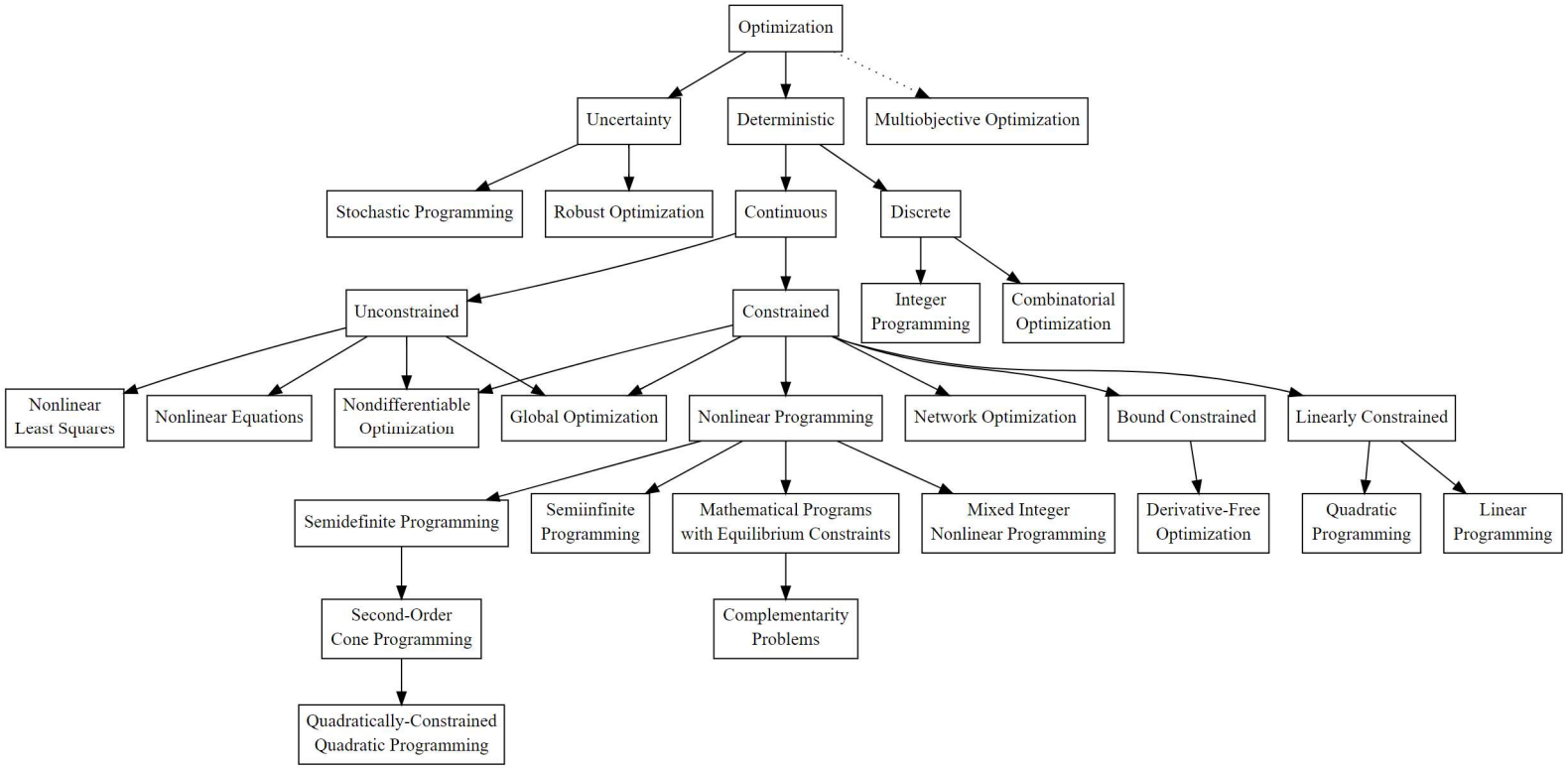
  - The Lagrange variables are now called dual variables and $d(\lambda, \upsilon)$ is concave in λ, υ
  - This dual function yields lower bounds on the optimal $f(x)$

$$d(\lambda, \nu) \leq p^*$$

- We can now form the Lagrange dual problem
  - What is the maximum lower bound that can be obtained from the dual function? Leads to certificate of optimality

$$\begin{aligned} \max \quad & d(\lambda, \nu) \\ s.t. \quad & \lambda \geq 0 \end{aligned}$$

Optimization Problem Taxonomy



Optimization Problem Types

- Linear Program (LP)
  - (P) Easy, fast to solve, convex

$$\begin{aligned} \min_{x \in X \subseteq \Re^n} \quad & f^T x \\ \text{s.t.} \quad & Ax \leq b \\ & A_{eq} x = b_{eq} \end{aligned}$$

  - Matlab command:
    - x = linprog(f, A, b, Aeq, beq, LB, UB, x0)
  - Python command:
    - scipy.optimize.linprog(f, A, b, Aeq, beq, bounds, x0)

Optimization Problem Types

- Quadratic Program (QP)
  - (P) Quadratic cost with linear constraints $O(n^3)$
    - Still fairly easy, fast to solve and convex

$$\min_{x \in X \subseteq \Re^n} \quad x^T Q x + c^T x$$
$$\text{s.t.} \quad Ax \leq b$$
$$A_{eq} x = b_{eq}$$

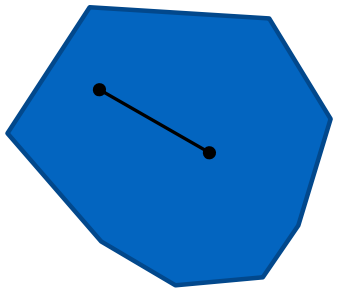
  - Matlab command:
    - x = quadprog(Q, A, b, c, Aeq, beq, LB, UB, x0)
  - Python command:
    - Go straight to scipy.optimize.minimize()

  - Examples with closed form solutions: Kalman filter, LQR

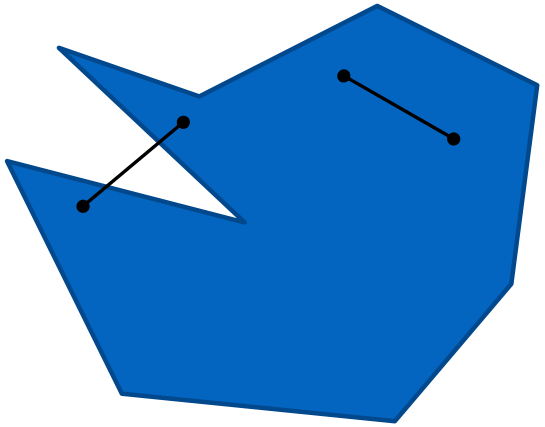
Convexity

- Definition: A set, *C*, is convex if any two points, x_1, x_2, in *C* can be connected by a line entirely in C.
  - That is, for all Θ in $[0,1]$, we have

$$\theta x_1 + (1-\theta)x_2 \in C$$



Convex

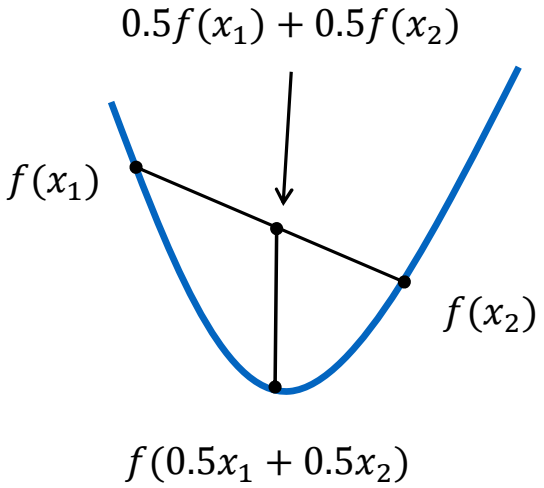


Nonconvex

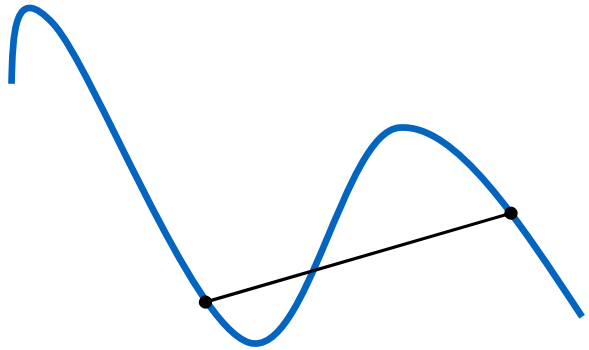
Convexity

- Definition: A function, $f(x)$, is convex if for any two points, x_1, x_2, and for all Θ in $[0,1]$, we have

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2)$$



Convex



Nonconvex

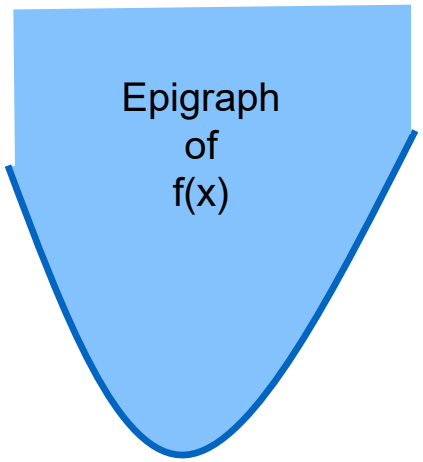
Convexity

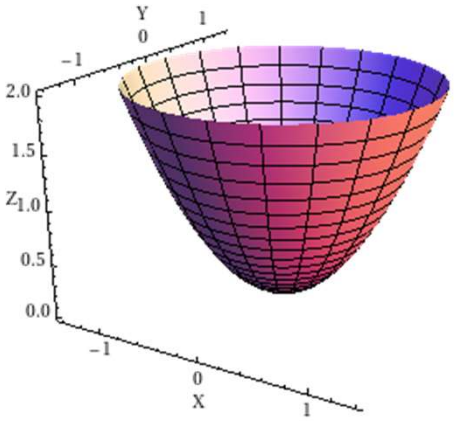
- A convex function has an epigraph that is a convex set

- Definition: A Convex Optimization problem is one where
  - $f(x)$ is a convex function
  - $g(x)$ is a convex function
  - $h(x)$ is an affine function

- This definition ensures the feasible region is a convex set
- Convex optimization problems have a unique global minimum!

- Common examples: linear programs, quadratic programs, geometric programs, second order cones and more





Optimization Problem Types

- Convex Program
  - (P) Convex problems are easy to solve, come with a certificate of optimality
  - $f(x)$ and $g(x)$ are convex functions, $h(x)$ is affine

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g(x) \leq 0 \\ & h(x) = 0 \end{aligned}$$

  - Good python solver: cvxopt, cvxpy, many others

Optimization Problem Types

- Non-Linear Program (NLP)
  - Non-convex problems are harder to solve, not guaranteed to find global optimum (local minima can occur)

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{s.t.} \quad & g(x) \leq 0 \\ & h(x) = 0 \end{aligned}$$

  - Matlab:
    - fmincon
  - Python
    - scipy.optimize.minimize, ipopt

Optimization Problem Types

- (Non-linear) Least Squares
  - Extremely common problem formulation with customized solution methods

$$\min_{x \in \mathbb{R}^n} \quad \frac{1}{2}||f(x)||^2$$

  - Linear least squares: $f(x) = Ax - b$
  - State estimation: $f(x_t) = y_t - h(x_t)$

  - Gauss-Newton and Levenberg-Marquardt solvers
    - Leaving for AER 1513

Optimization Problem Types

- Mixed Integer Linear Program (MILP)
  - (NP-hard) computational complexity

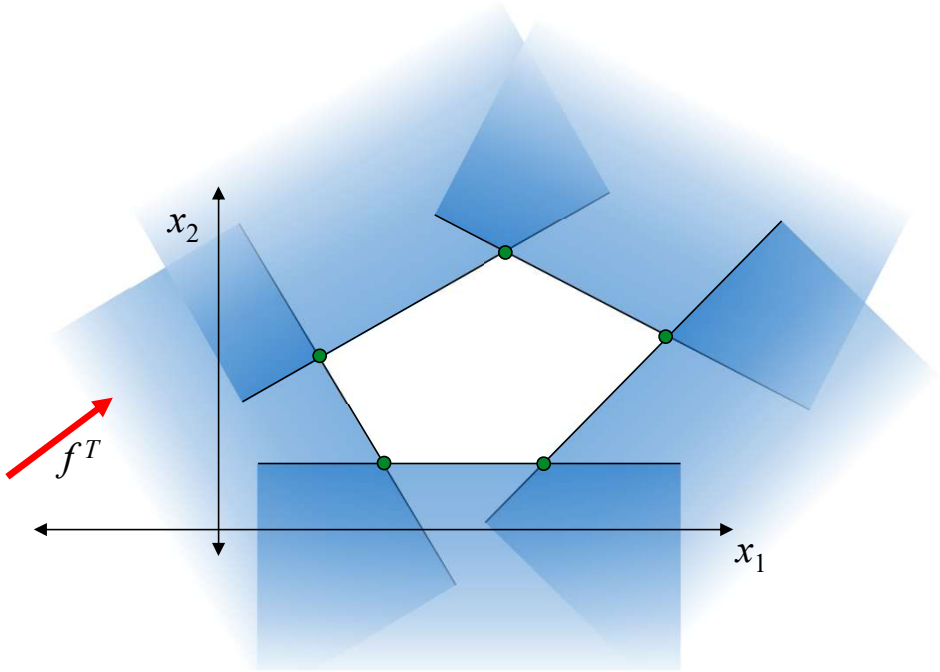
$$\begin{aligned} \min_{x \in X} \quad & f^T x \\ \text{s.t.} \quad & Ax \leq b \\ & A_{eq} = b_{eq} \end{aligned}$$

$$\text{where } X \subseteq Z^{n_i} \times \Re^{n_r}$$

  - Exponential growth in complexity
  - However, many problems can be solved surprisingly quickly
    - CPLEX, Gurobi
- MINLP, MILQP etc.

Solution Methods for Linear Programs

- Simplex Method
  - For a linear program, optimum must be at the intersection of constraints
  - Intersections are easy to find, change inequalities to equalities, add slack variables
  - Jump from one vertex to the next (in a smart way), until no more improvement is possible

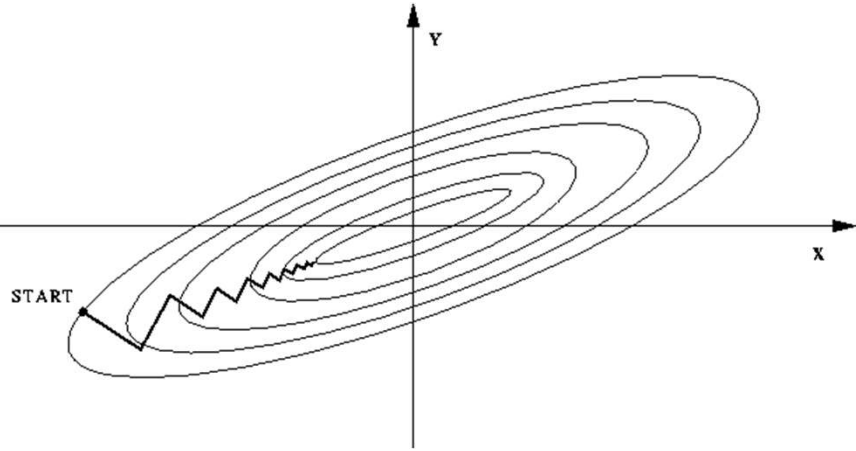


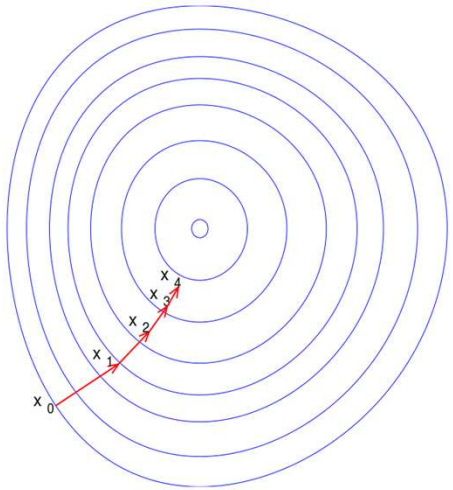
Generic Solvers - Gradient Descent

- At any point, the negative gradient vector defines a direction of steepest descent locally
  - Given a feasible solution x_k, there exists $\alpha > 0$ such that

$$f(x_{k+1}) \leq f(x_k) \qquad \text{for} \qquad x_{k+1} = x_k + \alpha \nabla f(x)$$

  - The step size or learning rate (α) can be defined in advance, or found via line search
    - Constant step size
    - Diminishing step size
    - Exact and inexact line search
  - Termination based on gradient magnitude (NC check)



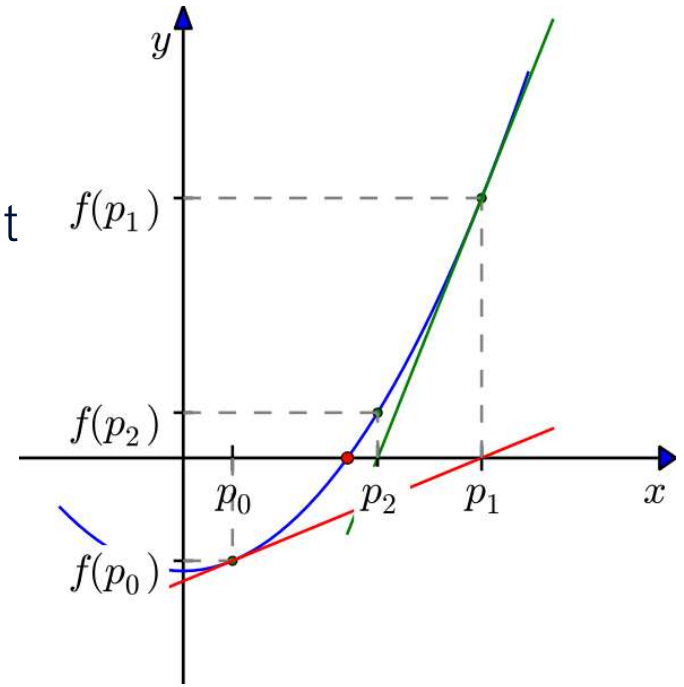


Newton-Raphson Method

- Based on the second order Taylor series approximation instead of the first for gradient

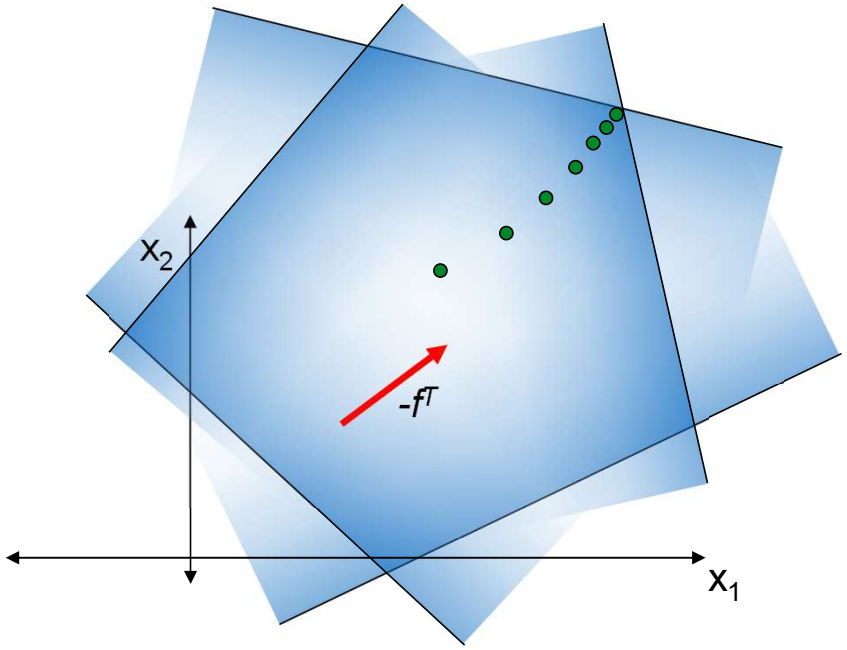
$$x_{k+1} = x_k + \alpha(\nabla^2 f(x_k))^{-1}\nabla f(x)$$

- Jumps to the solution of the quadratic approximation t
- Many extensions, and customizations



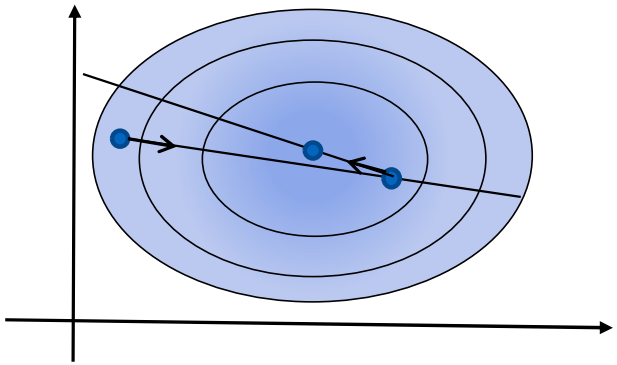
Solution Method for Nonlinear Programs

- Interior Point Methods
  - Apply Barrier Function to each constraint and sum
  - Primal-Dual Formulation
  - Newton Step
  - At each iteration,
  - increase slope of barriers
- Benefits
  - Applicable to all convex problems
  - Scales better than Simplex for LP
  - Certificate of Optimality
    - Stop whenever
    - Know how close to optimal the current solution is
    - Relies on duality



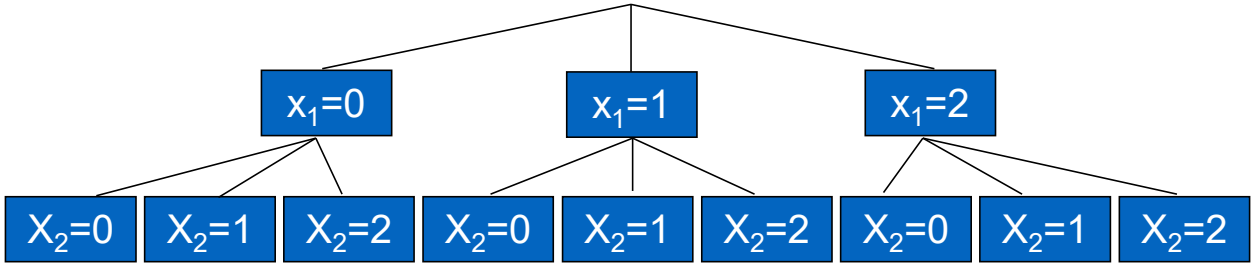
Solution Methods For Nonlinear Programs

- Sequential Quadratic Programming
  - Also an interior point method
  - At each iteration, calculate gradient and Hessian of Lagrangian
  - Use Newton step direction as a descent direction and apply a line search
  - Finding Newton step involves inverse of Hessian
    - Can be approximated through diagonal, BFGS, delayed updates



Solution Methods for Integer Programs

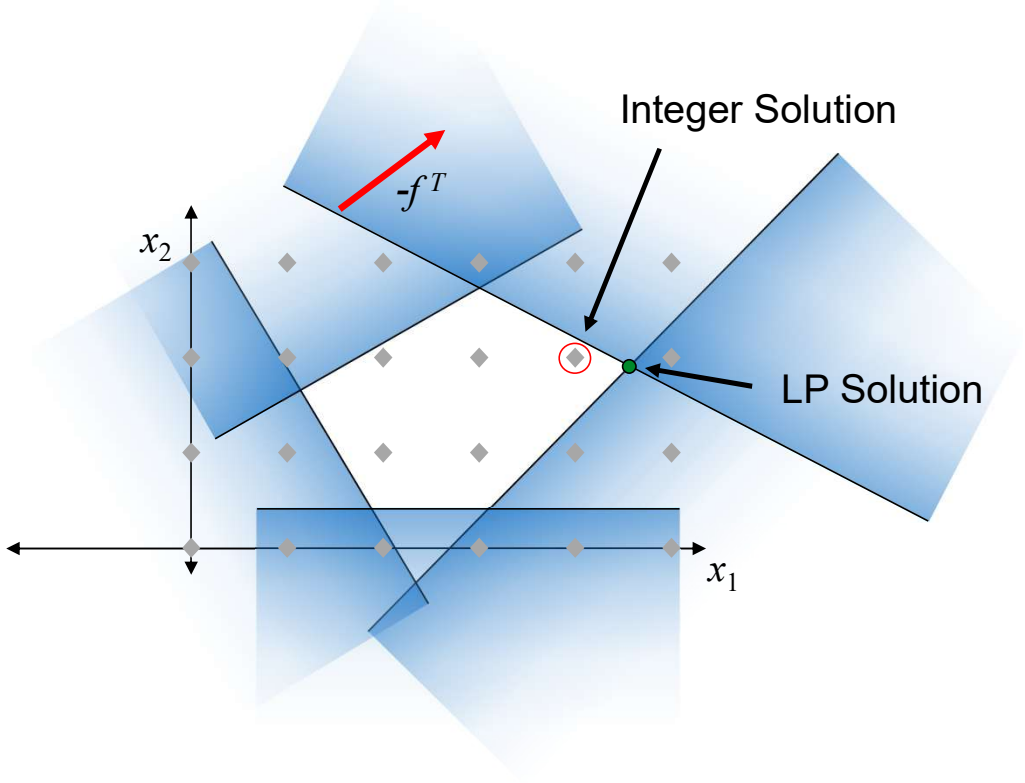
- Enumeration – Tree Search, Dynamic Programming etc.



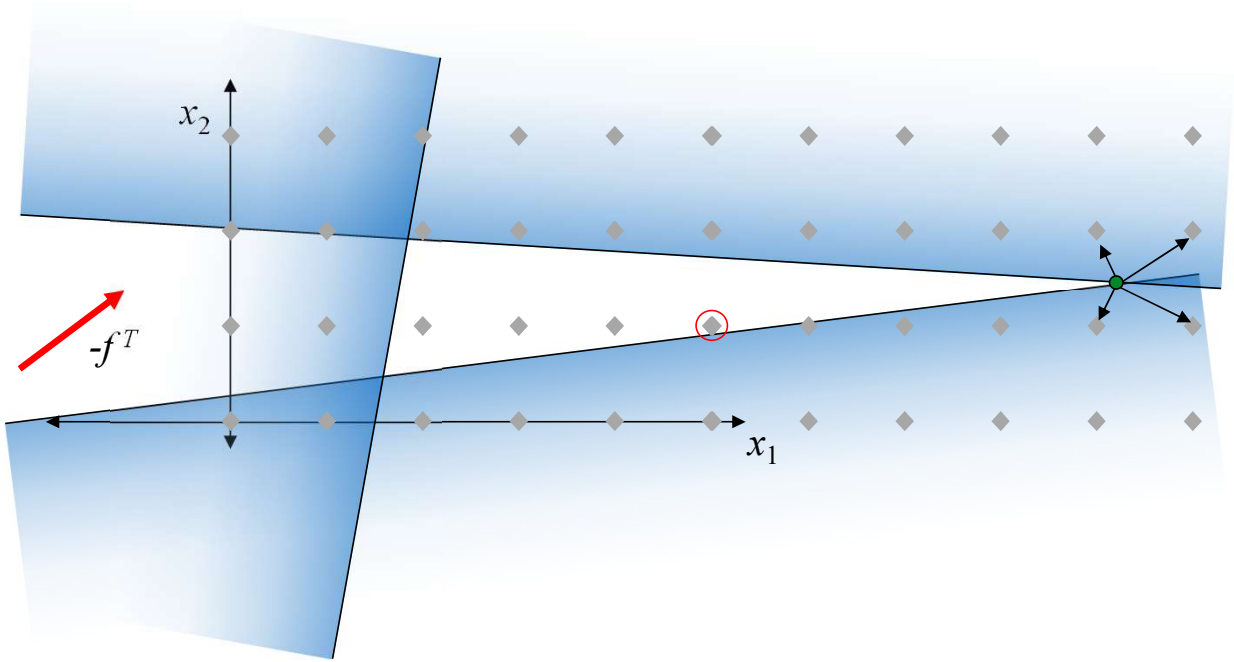
- Guaranteed to find a feasible solution (only consider integers, can check feasibility (P))
- But, guaranteed exponential growth in computation time

Solution Methods for Integer Programs

- How about solving LP Relaxation followed by rounding?



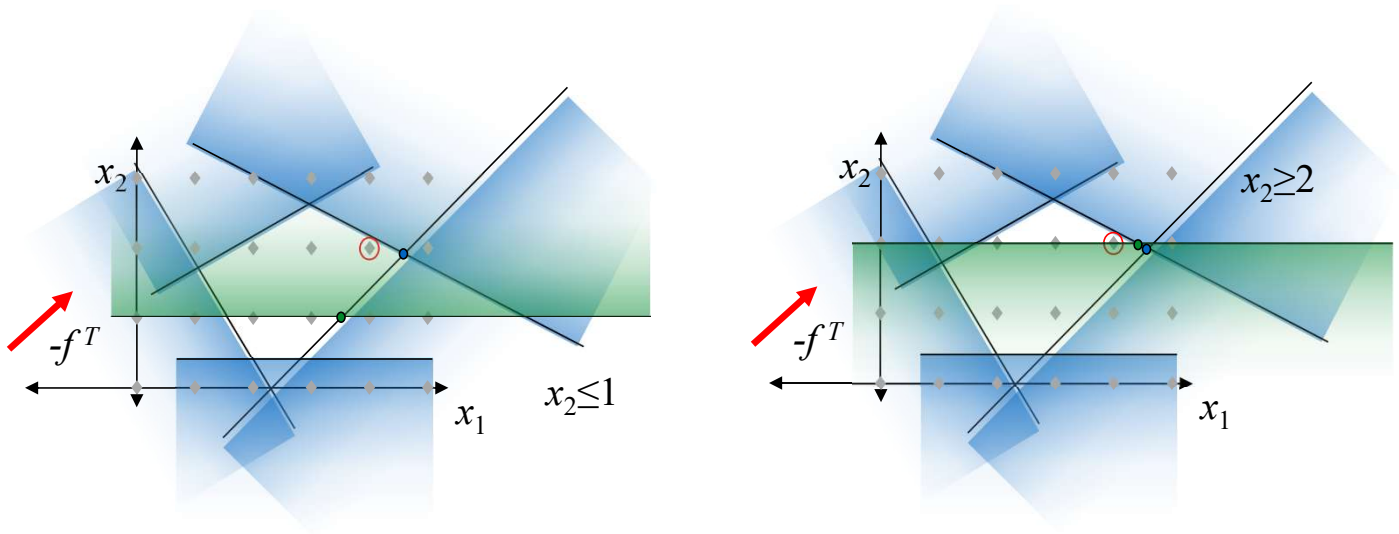
Integer Programs



- LP solution provides lower bound on IP
- But, rounding can be arbitrarily far away from integer solution

Combined approach to Integer Programming

- Why not combine both approaches!
  - Solve LP Relaxation to get fractional solutions
  - Create two sub-branches by adding constraints

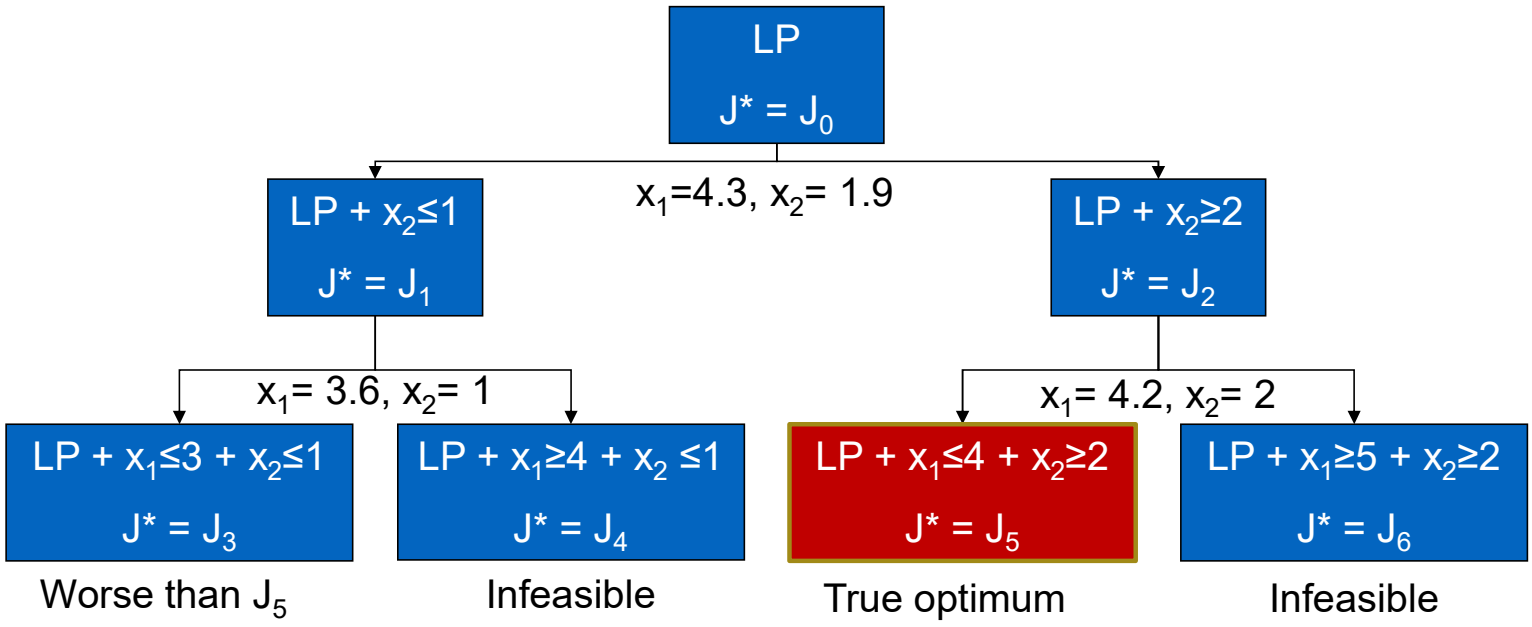


Branch and Bound Method

- Solve LP relaxation for lower bound on cost for current branch
  - If solution exceeds upper bound, branch is terminated
  - If solution is integer, replace upper bound on cost if lower
- Create two branched problems by adding constraints to original problem
  - Select integer variable with fractional LP solution
  - Add integer constraints to the original LP
- Repeat until no branches remain, return optimal solution.

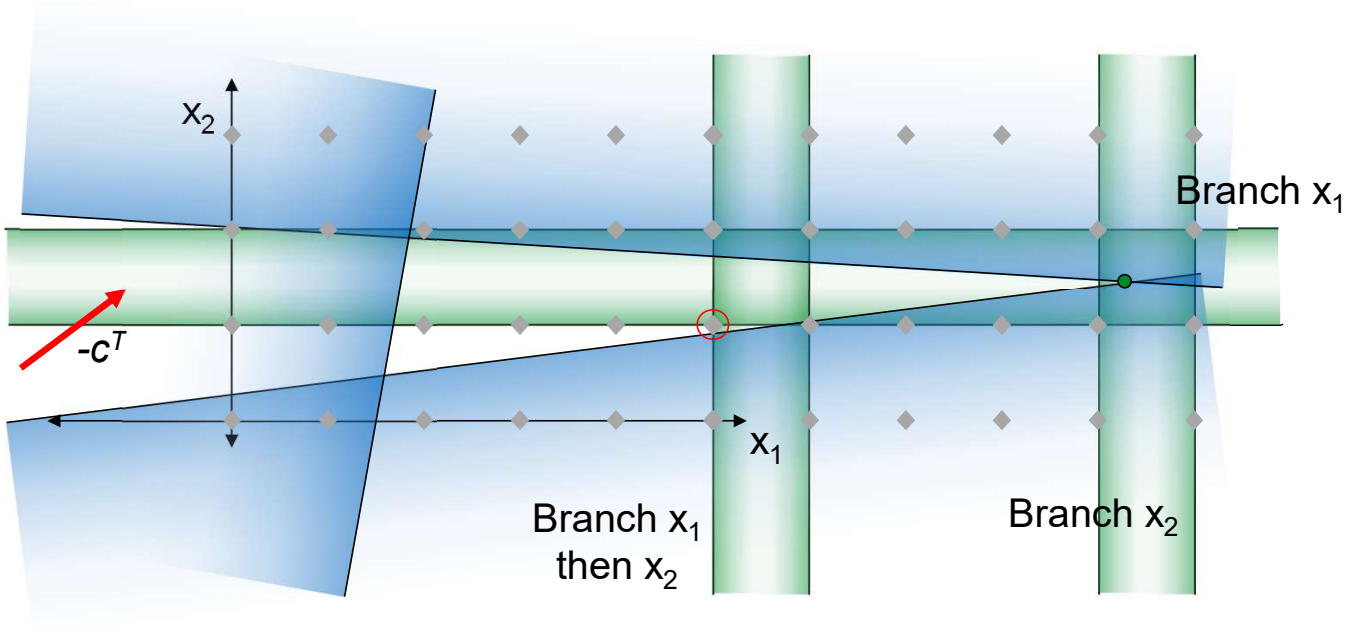
Solution Methods for Integer Programs

- Branch and Bound
  - Branch as above
  - LP give lower bound, feasible solutions give upper bound



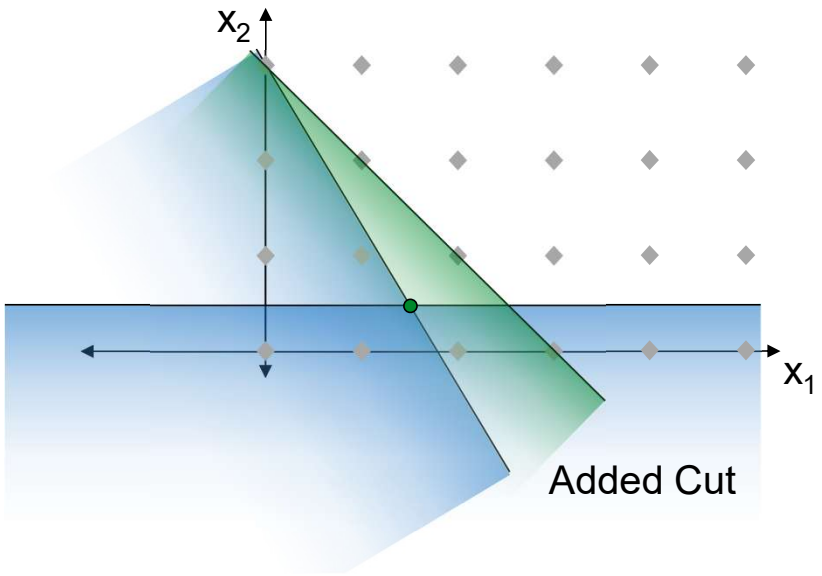
Integer Programs

- Order matters
  - All solutions cause branching to stop
  - Each feasible solution is an upper bound on optimal cost, allowing elimination of nodes



Additional Refinements –Cutting Planes

- Idea stems from adding additional constraints to LP to improve tightness of relaxation
- Combine constraints to eliminate non-integer solutions



- All feasible integer solutions remain feasible
- Current LP solution is not feasible

Summary

- Optimization involves identifying solutions that meet necessary and sufficient conditions for optimality
  - Often only possible to identify locally optimal solution
- Problem complexity depends on characteristics of cost, constraints, domain
  - Huge taxonomy of different problems, each with their own solution methods
- Convex/non-convex the main threshold between easy and hard, then differentiable/non-differentiable
- A wide array of solution methods exist, easily employed through standard software packages
- As problem size increases as in deep learning, reversion to gradient methods necessary
  - Memory limitation becomes primary factor in selection of search method